Jashore University of Science and Technology

Jashore-7408, Bangladesh

Report on Internship

at Unicorn Software & Solutions Ltd (USSBD)

Submitted By:

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TO WHOM IT MAY CONCERN

This is to certify that Rakesh Biswas (ID:200112), a student of Jeshore University of Science and Technology, has successfully completed his internship program at Unicorn Software and Solutions from 15/01/2026 to 15/04/2026. During this period, he was assigned to the Web Application Developer and worked under the Company supervision of Md Harun Or Rashid.

During his internship, Rakesh Biswas demonstrated strong technical ability, dedication, and a genuine enthusiasm for learning. He actively contributed to various development tasks under the supervision of our Senior Developer and quickly adapted to our workflow and team environment.

His problem-solving mindset, teamwork, and professionalism were commendable. Rakesh showed great initiative in understanding new technologies and completing his responsibilities with accuracy and efficiency.

We believe he has the potential to grow into a skilled software engineer, and we wish him continued success in his future career.

Sincerely,

Abu Bakar Siddique
CEO and Managing Director
Unicorn Software and Solutions
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Dedication

I want to dedicate this work to my parents for their endless love, support and encouragement throughout my academic journey and internship.

Acknowledgments

I am really grateful to God above all for granting me the strength, patience and determination to complete the internship. I want to expand my heartfelt thanks to my supervisor- Md. Harun Or Rashid, Senior Software Developer at Unicorn Software Solutions Ltd(USSBD), for his continuous guidance and support throughout the development of the VerbaSense project.

I am also thankful to the faculty members of the Department of Computer Science and Engineering, Jashore University of Science and Technology , for their academic support. I am forever grateful to my parents and family for their endless love, sacrifices and moral encouragement without which this venture would not have been feasible.

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Abstract

This report shows a detailed overview of the design and development of **VerbaSense**. Which is a high-performance Speech to-Text (STT) application developed during my internship at Unicorn Software & Solutions Ltd (USSBD). VerbaSense leverages the OpenAI Whisper engine to provide secure, local and accurate audio transcriptions.

The project was built using a modern full-stack architecture. That includes Next.js 14 for the frontend and Node.js/Express for the backend. Key features include : real-time transcription, history management and secure JSON Web Token (JWT)-based authentication. This report covers the methodology, system architecture, implementation details and evaluation of the platform.

Chapter One

Introduction

Chapter One: Introduction

In the rapidly changing context of **Software Engineering** and **Artificial Intelligence (AI)**, there should be rigorous practical application alongside theoretical expertise to develop industry-standard solutions. As global industries undergo a huge digital-shift, academic institutions are prioritizing internships to bridge the gap between classroom algorithms and production ready deployments. For a **Computer Science and Engineering** student(Undergraduate), an internship at a specialized software firm offers an unparalleled scope-to understand the complexities and to learn of the **Software Development Life Cycle (SDLC)** and the nuances of integrating advanced AI engines into user centric platforms.

This report gives a detailed reflection on my professional journey as a **Software Engineering Intern** at- **Unicorn Software & Solutions Ltd (USSBD)**. Since its begin in 2005, **Unicorn Software & Solutions Ltd (USSBD)** has been at the forefront of the local IT sector. Which is specialized in custom enterprise applications and innovative web architectures. My working period at the Khulna branch was centered on the development of **VerbaSense**—a high-performance **Speech-to-Text (STT)** application -which was designed to bring the power of the **OpenAI Whisper** engine to a secure, locally-hosted environment.

The initial purpose of this report is to analyze the technical architecture of **VerbaSense**, to document the development methodologies employed and also to evaluate the performance of the integrated AI models. Moreover, it highlights the professional growth and technical proficiency gained during this tenure -which provides a technical roadmap for building scalable AI-integrated web applications.

1.1 Organizational Overview: USSBD

Unicorn Software & Solutions Ltd (Unicorn Software & Solutions Ltd (USSBD)) stands as a leading software development and IT consulting company, which headquartered in **Dhaka, Bangladesh**. Since its begin, the company has dedicated itself to transforming the digital landscape of the region by providing state-of-the-art software solutions that are both scalable and secure. With nearly **2-decades** of operational excellence, **USSBD** has built a reputation for ex-

cellence in delivering **high performance applications** -which serve customers worldwide. The organization is driven by a team of highly skilled engineers, designers and project managers who utilize **modern development methodologies** to solve complex industrial problems.

1.1.1 Historical Background and Formation

Unicorn Software and Solutions Limited (USSBD) began its journey in **2005**, when digital economy of Bangladesh was in its very early phase. The company was founded by a visionary group of **IT-professionals** who recognized the critical need for a reliable technology partner that could deliver more than just basic web presence. Over the last **twenty years**, the company has undergone important evolution and has transitioned from a **small development studio** into a **mature software house** with strong infrastructure. This journey has been marked by the successful delivery of over **45 high impact projects** and ranging from retail management systems to **sophisticated AI-driven platforms**. Now **USSBD** is familiar as a key contributor to the nation's **"Digital Bangladesh"** vision- who is continuously pushing the boundaries of what is possible in the local IT sector.

1.1.2 Organizational Vision and Strategic Goals

USSBD's main goal is to be a key driver of **digital transformation** in the region and empower organizations through **intelligent and human centric technology**. The mission of the company is based on the belief that software should not only be functional, but also give a **competitive advantage** to its users. To gain this, **USSBD** has set several key strategic goals:

- **AI-driven product innovation:** The company is committed to research and development of its own tools. That includes- **Large Language Models (LLMs)** and **advanced AI engines**. Through focusing on projects like **Verbasense**, **USSBD** targets -to make **cutting-edge AI technology** accessible and secure for enterprise use.

- **Scalable Enterprise Architecture:** A key strategic goal is to design a **software back-**

bone which is built for growth. It includes using **microservices architecture**, **cloud-native deployments** and **robust database management systems**, so that products can be **seamlessly scaled** as client needs to grow.

- **Human-Centered Design (HCD): USSBD** gives priority to **user experience** by adopting a **design-first approach**. This ensures that even the most complex backend functionalities are presented through an **intuitive and accessible user interface**. Which is really making the learning process easier for clients.

- **Digital Readiness and Consulting:** In addition to development, the company has aim to increase the **digital literacy and readiness** of its customers by providing **comprehensive IT consulting and support services**.

1.1.3 Core Services and Offerings

Unicorn Software & Solutions Ltd (USSBD) offers a variety of professional software services specifically designed to fill the needs of the global market:

- **Custom Enterprise Solutions:** The company is specialized in building customized management systems, including ERP (Enterprise Resource Planning), CRM (Customer Relationship Management) and specialized retail management tools. Those solutions are designed to automate core business processes and to improve operational efficiency.

- **Full-stack web architecture:** Using modern frameworks like:- Next.js, React and Node.js, USSBD builds high performance web platforms that are optimized for SEO, accessibility and speed. These architectures are designed to maintain high concurrency and give smooth user experience.

- **Mobile Ecosystem:** The development team is expert at building native and cross-platform mobile applications (Flutter, React Native) for both iOS and Android. These apps focus on giving- seamless transition from desktop to mobile for enterprise users.

- **AI and Machine Learning Integration:** A core distinguishing feature of USSBD is its ability to incorporate the advanced AI models into existing software workflows. These include speech-to-text, natural language processing (NLP) and predictive analytics tools.

- **Secure IT Infrastructure:** The company gives enterprise grade design and secure hosting solutions. This keep client data safe with industry standard encryption and security protocols.

1.1.4 Geographic and Physical Presence

To ensure high levels of engagement and service delivery with customers, **USSBD** operates from two strategic development centers in Bangladesh:

- **Dhaka Headquarters:** Located in the heart of **Uttara**, this office serves as the administrative center and the main hub for design and customer relations. It is equipped with **modern infrastructure** to support a large development workforce.

- **Khulna Development Branch:** Located in **Nirala R/A, Khulna**. This branch is the **core research and development center** where much of the technical implementation and AI research takes place. My internship was conducted at this location. Which provides me with **direct exposure** to the core development team.

1.1.5 International Linkages and Collaborations

USSBD is committed to maintaining **international standards** in software quality, documentation and security. The company fully active in collaborating with **global technology partners** to stay abreast of emerging trends in the global IT sector. By adopting **internationally recognized development standards** (such as **ISO guidelines** for software quality), **USSBD** ensures that-its products are **globally competitive**. This international perspective is reflected in the company's ability and expertiseness to serve a variety range of clients through different continents. That provides them with **future-proof digital solutions**.

In conclusion **USSBD** shows a modern, **forward-thinking software firm** that prioritizes **innovation and engineering excellence**. Understanding the organization's history, vision and core services provides the necessary context for the work I performed during my **office-period**. With the solid understanding of the company's operational framework, next chapter will detail the specific structure and objectives of my internship program.

Chapter Two

Internship Structure and Objectives

Chapter Two: Internship Structure and Objectives

2.1 Overview of the Internship Program

USSBD's internship program has been carefully structured to give final year undergraduate students with a **comprehensive and intensive introduction** to the professional world of **software engineering**, **systems architecture** and **large-scale project implementation**. My stint at the **Khulna branch** was not just a passive observation period. Rather it was a well thought out **12-week program** -which was designed to bridge the gap between the institutional theory of **Jessore University of Science and Technology (JUST)** and the increasing demands of the **global IT industry**. From the very first week, the program emphasized a **"learning by doing"** philosophy. Where I was integrated into the **core development team** and given **real responsibilities** under the **supervision of senior developers**.

Unlike many typical internships, where students are often limited to menial administrative tasks. In this case- the **USSBD framework** emphasizes **active participation** and **professional mentoring**. I was really encouraged to rotate through different stages of the **development lifecycle**—from **initial requirements gathering** and **UI/UX prototyping** to **backend API design** and **AI engine optimization**. This rotation ensured a **combined experience**—which is allowing me to understand how the various components of a complex application like **VerbaSense** come together to create a **seamless user experience**. The mentorship I received included: **regular technical briefings**, **in-depth code reviews** and meetings, those created a **fast-paced environment**.

2.2 Specific Internship Modules

To ensure structured technical growth and a deep understanding of modern web architectures, the internship was divided into two key technical modules, each focusing on a distinct layer of the VerbaSense ecosystem:

Module 1: Full-Stack Web Development (Frontend & Backend)

Location: Khulna Branch Office, Nirala R/A

- **Duration:** Approximately 7 weeks
- **Objective:** Master the end-to-end development of modern, scalable web applications using the Next.js and Express ecosystems while adhering to industry-standard security and performance benchmarks.
- **Key Learning Tools:** Next.js 14 (App Router), Node.js, Express.js, MongoDB, Tailwind CSS, Postman and GitHub.
- **Learning Activities:**
  - **Responsive UI Design:** Designing and implementing fluid, mobile-first user interfaces using Tailwind CSS and Next.js Server Components to ensure optimal load times and SEO performance.
  - **API Architecture:** Developing robust RESTful API endpoints for handling secure user authentication (JWT), file upload streams and real-time state management on the backend.
  - **Database Management:** Implementing and optimizing MongoDB schemas for the persistent storage of complex transcription metadata and user history, focusing on data integrity and query efficiency.
  - **Testing and Debugging:** Performing rigorous end-to-end testing of the application flow using tools like Postman to identify and resolve performance bottlenecks in the API layer.

Module 2: AI Engine Integration and Audio Processing

Location: Khulna Branch Office, Nirala R/A

- **Duration:** Approximately 5 weeks

- **Objective:** Gain deep technical expertise in the deployment of locally-hosted AI models and the complexities of handling high-performance audio processing in a web environment.
- **Key Learning Tools:** OpenAI Whisper (C++ CLI), FFmpeg, Node.js Child Processes and system-level resource monitoring tools.
- **Learning Activities:**
  - **AI Model Deployment:** Researching, deploying and fine-tuning the OpenAI Whisper engine for local inference, ensuring the system can handle transcription tasks without relying on external cloud APIs.
  - **Audio Pre-processing:** Developing automated scripts using FFmpeg to normalize, trim and convert diverse audio formats (MP3, WAV, FLAC) into optimized streams for the transcription engine.
  - **Resource Optimization:** Analyzing and optimizing the CPU and RAM allocation for concurrent AI inference tasks to ensure the server remains responsive during heavy transcription loads.
  - **Secure Integration:** Designing stateless authentication layers and secure file-handling protocols to ensure that user-uploaded audio remains private and secure during the transcription process.

2.3 Work Ethic

During the course of this internship, maintaining a strong and consistent work ethic was necessary not only for my individual professional growth but also for contributing meaningfully to the high stake development goals at USSBD. The professional environment into the company is demanded a level of consistency, discipline and accountability—which went far beyond the expectations of a typical, normal classroom settings.

- **Punctuality and Commitment:** I tried to strictly maintain the time during my professional office schedule, like—arriving on time for timely meet-up and technical briefings. And thus the consistency really helped me to create disciplined mindset and demonstrated my reliability to my mentor.

- **Accountability and Ownership:** Whenver I was tasked with a critical API fixing or researching the intricacies of the Whisper engine, I took full ownership of my assignments. I learned that in a professional environment, being accountable for one's code—including the bugs—is the first step toward becoming a senior-level engineer.

- **Attention to Detail:** Actually production coding requires a careful and structure approach. I developed a habit of double-checking my logic, variable naming conventions and API response structures to reduce avoidable errors. This attention to detail ensured that, the VerbaSense platform was stable and ready for production.

- **Self-Initiative and Research:** In a rapidly changing industry, actually the direct guidance is not always possible to get. I actively researched technical hurdles, spending a lot of hours reading official documentation for Next.js, Node.js and FFmpeg. Sometimes, I saw official video for understanding the setup or concept. This step not only solved immediate issues but also importantly extended my technical self-trust.

- **Consistency in Learning:** I maintained regular technical log where I carefully checked new concepts, challenges faced and solutions implemented. This practice of regular review allowed me to build on my knowledge progressively. This also ensures that I never repeated the same mistake twice.

- **Professional Integrity and Respect:** During the collaborative sessions and code reviews, I respected the talents of senior developers and tried to their approached feedback with humble, learning oriented mindset. That cooperative spirit really helped me to incorporate into the company's culture and build lasting professional relationships.

The internship structure at USSBD gave perfectly balanced mix of theoretical research and high-pressure implementation. Through working these rigorous modules and adhering to a professional work ethic, I was able to prepare myself for the technical challenges detailed in the following chapters. This experience has not only enhance my technical proficiency but has also shaped my understanding of what it means to be a professional software engineer in the modern era.

Chapter Three

My Involvement in Office Work

Chapter Three: My Involvement in Office Work

3.1 Internship Goals and Deliverables

The primary purpose of my internship at **USSBD** was to develop the **Verbasense (speech-to-text platform)** from start to finish. This project was created to meet up the increasing demand for a highly accurate, private transcription service—which does not depend on expensive and potentially insecure cloud-based APIs. My responsibilities included overseeing the whole project lifecycle—from initial feasibility studies of AI models to deployment of a fully functional production-grade application. The key goals set during the first phase of my tenure were not only functional requirements, but also benchmarks of **engineering excellence**:

- **High-Performance Frontend:** A responsive, mobile-first user interface built with Next.js 14 and Tailwind CSS, focusing on minimal bundle sizes and sub-second load times.
- **Stateless Backend API:** A robust and scalable server architecture using Node.js and Express, designed to handle high-concurrency audio processing without state-related bottlenecks.
- **Local AI Integration:** A seamless integration of the OpenAI Whisper engine, optimized for execution on local hardware to ensure absolute data sovereignty.
- **Security and Integrity:** Implementation of a robust authentication system using JWT and secure database schemas for managing user transcription history and metadata.
- **Documentation and Version Control:** A maintainable, well-commented codebase stored in GitHub, accompanied by comprehensive technical documentation for future scalability.

3.2 USSBD Development Hub – Khulna Branch

Basically the majority of my professional involvement has revolved around the Khulna branch of USSBD. That center serves as the organization's key- research and development center, housing specialized teams responsible for AI integration and full-stack software architecture.

3.2.1 Physical Infrastructure Overview

The physical infrastructure at the Khulna branch is carefully designed to support high intensity software engineering. My dedicated workstation featured high-performance computing hardware. Which were essential for running local AI inference tasks. The environment encouraged a culture of direct mentoring—where I had immediate access to my supervisor whenever technical obstacles arose. And this was allowing for rapid problem-solving and architectural thinking.

3.2.2 Server Configuration and Resource Management

Running AI models like Whisper on a local computer needs a lot of careful planning. In the first few weeks, I worked closely with my supervisor to learn how our servers were set up to handle multiple transcriptions at once without slowing down or crashing.

- **GPU and Memory (VRAM):** Since AI works best with graphics cards (GPUs), I learned how to monitor the graphics memory. This was important to make sure we weren't trying to run too many transcription tasks at once, which could run out of memory and stop the service.

- **Processing Power (CPU):** For tasks like cleaning up and fixing audio files before they are transcribed, the main processor (CPU) does a lot of work. I learned how to manage these tasks so the computer didn't get too hot or slow down other important office work.

- **Handling Large Files:** When someone uploads a very large audio file, it can easily use up all the available memory. I learned how to process the data in small pieces (streaming) so the system stayed stable even when dealing with very long recordings.

3.2.3 Development Network and Secure Access

Data security and user privacy are paramount concerns at USSBD. The development network was logically and physically segmented to provide a secure environment for testing the Ver-

baSense platform.

- **Dedicated Research Workspace:** I worked in a dedicated research setup that was kept separate from the rest of the office's main systems. This was very important because testing heavy AI models and processing large files needs a lot of power, and this separation made sure my work didn't slow down or disturb other employees in the branch.

- **Secure Data Handling Protocols:** I learned to implement encrypted communication channels (HTTPS) for all interactions between the web server, the transcription engine and the MongoDB instances. This taught me the importance of securing "data in transit."

- **Access Control and Authentication:** Following the "Least Privilege" principle, I was granted access only to the specific servers and databases required for the project. This exposure to professional access control provided me with a solid understanding of how USSBD protects its intellectual property and user data.

3.2.4 Agile Workflow and Task Management

To maintain a high project velocity, I followed a streamlined Agile workflow focused on direct collaboration with my supervisor. This structured approach was instrumental in my transition from a student mindset to that of a professional software engineer.

- **Regular Task Updates and Progress Tracking:** I maintained a consistent communication loop with my supervisor, providing detailed updates on my task progress every day or two. These informal yet structured updates allowed me to discuss technical hurdles as they arose and ensure that my development work remained perfectly aligned with the overall project roadmap. This process helped me refine my ability to explain complex engineering challenges in a concise and professional manner.

- **Iterative Planning:** I learned how to break down the massive VerbaSense roadmap into granular, actionable tasks. By focusing on short-term milestones, I was able to deliver

functional "mini-releases," such as a working file-upload layer or a basic transcription history dashboard, which allowed for constant feedback from my mentor.

- **Technical Reviews and Collaborative Coding:** Every major code contribution I made was subject to a thorough review by my supervisor. Discussing my architectural decisions with a senior expert taught me the importance of code readability, maintainability and the value of professional documentation in a production environment.

3.3 USSBD Corporate Coordination and Office Engagement

This section highlights the organizational and collaborative aspects of my internship, focusing on how I integrated into the USSBD office environment and coordinated with my supervisor to achieve the project's strategic goals. Beyond the technical code, these interactions were essential for understanding how enterprise-level software is conceived, planned and managed within a professional corporate framework.

3.3.1 Technical Strategy Sessions with Supervisor

A significant portion of my office work involved intensive technical strategy sessions with my supervisor. These meetings were far more than simple status updates; they were structured, deep-dive discussions into the architectural future of the **VerbaSense** platform. Often held over a whiteboard, we spent hours evaluating different AI models and discussing the complex trade-offs between system latency, transcription accuracy and VRAM consumption. One of the most critical decisions we reached together was the move from a pure Python-based implementation to a specialized C++ bridge for the Whisper engine, which significantly improved the application's performance on local hardware. These sessions taught me how to present technical arguments effectively, defend my architectural choices and pivot my development strategy based on high-level feedback. The direct mentorship I received was instrumental in shaping my understanding of the "big picture" in enterprise software design.

3.3.2 Collaborative Resource Planning and Decision Making

Software development in a high-intensity environment like USSBD requires meticulous resource planning. I coordinated closely with my supervisor to identify and secure the hardware and software resources necessary for the project's success. This involved formalizing requests for dedicated GPU compute time and discussing the allocation of system memory for our local staging environment. We also collaborated on designing the database schema, weighing the benefits of NoSQL denormalization against data consistency requirements. Participating in these decision-making processes gave me a unique insight into how professional IT departments manage their technical assets and budgets to ensure that development pipelines remain uninterrupted without overextending the branch's local infrastructure.

3.3.3 Office Culture and Professional Communication

Integrating into the professional office culture at the Khulna branch was a vital part of my growth as an engineer. I learned to navigate the nuances of corporate communication, transitioning from informal student-level interactions to drafting formal progress reports and engaging in high-level technical discourse with my supervisor. I adhered strictly to the office's professional ethics, maintaining a disciplined schedule that mirrored the organizational rhythm. I also learned the value of professional documentation; my mentor emphasized that "code is only as good as its documentation," which led me to prioritize writing clear, concise README files and technical logs. This immersive experience helped me develop the "soft skills" necessary for a successful career, such as technical empathy, active listening and the ability to work effectively within a high-output professional team.

3.4 Technical Implementation and Development Labs

This section details the core engineering work and the series of technical experiments conducted to optimize the VerbaSense platform. These "labs" were essential for making data-driven deci-

sions regarding the application's architecture and performance.

3.4.1 Core Implementation: OpenAI Whisper and API Layer

The implementation of the VerbaSense backend required a robust bridge between a modern web environment and a high-performance C++ AI engine. My primary focus was on building a resilient and scalable API layer.

- **Backend Bridge and Process Management:** I utilized the **Node.js child_process** module to spawn the Whisper engine as a background worker. This decoupled approach allowed the main Express.js server to remain non-blocking, ensuring that user requests (such as UI updates or history fetching) were handled while the AI engine performed heavy inference in the background.

- **Asynchronous Stream Handling:** I implemented a real-time output parser that captured the engine's stdout. This allowed us to extract the transcribed text and its associated timestamps dynamically, which were then structured into a JSON format for delivery to the frontend.

- **Frontend-Backend Communication:** The communication layer was built using RESTful principles, with a specific focus on handling multi-part form data. I developed custom middleware to manage audio file uploads, ensuring that temporary files were sanitized and deleted immediately after processing to maintain server security and disk integrity.

3.4.2 Lab 1: Audio Processing and Normalization Benchmarking

In this lab, I focused on making sure the audio files were in the best possible format before sending them to the transcription engine. I tested different types of audio—like high-quality music files, standard mono recordings and low-quality phone recordings. I noticed that very high-quality files took too much time to process without actually making the text any better, while low-quality files caused the AI to make a lot of mistakes. After some testing, I found

that converting everything to a 16kHz mono format using FFmpeg was the perfect balance. It made the transcriptions much faster and more reliable, regardless of how the user recorded their audio.

3.4.3 Lab 2: Inference Speed and Memory Optimization

For the second lab, I needed to figure out which AI model would work best on our office computers. I tested the Tiny, Base and Small versions of the Whisper model to see how they handled speed and computer memory. The Tiny model was very fast but not accurate enough and the Small model was accurate but used up almost all the computer's memory, which would slow everything else down. I found that the Base model was the best choice—it was accurate enough for professional use but didn't crash the system. Because of this, my supervisor and I decided to use the Base model for VerbaSense, ensuring it runs smoothly for everyone.

3.5 Tools and Technologies Used

The development of the VerbaSense platform relied on a modern, robust and highly integrated tech stack. This section details the tools and technologies selected to meet the project's intensive computational and security requirements, as specified in the official project documentation.

3.5.1 Frontend Development Stack

To deliver a responsive and high-performance user experience, the frontend was built using the **Next.js** framework. This choice allowed for efficient server-side rendering and optimized page loads.

- **TailwindCSS and Shadcn UI:** For styling, I utilized TailwindCSS for its utility-first approach, combined with Shadcn UI for accessible and aesthetically pleasing components.
- **Framer Motion:** To enhance user engagement, I implemented smooth UI transitions and micro-animations using Framer Motion, giving the platform a premium feel.

- **Web Audio API:** I integrated the Web Audio API to enable direct, high-fidelity microphone recording within the browser, facilitating a seamless user-to-server audio pipeline.

3.5.2 Backend and Security Framework

The backend was architected using **Node.js** and **Express.js**, implemented with **TypeScript** to ensure type safety and long-term maintainability.

- **Authentication and Security:** User security was prioritized through the implementation of **JSON Web Tokens (JWT)** for session management and **Bcrypt** for robust password hashing.
- **Audio Stream Management:** I used the **Multer** middleware to handle the secure upload and temporary storage of '.wav' files, ensuring that the backend remained efficient even during high-concurrency transcription tasks.

3.5.3 AI Inference and Audio Engine

The core of the application is the **OpenAI Whisper** engine, which was integrated to provide industry-leading speech-to-text accuracy.

- **Whisper.cpp and FFmpeg:** I utilized the C++ wrapper for Whisper to enable local, high-performance inference. **FFmpeg** served as the essential audio "sanitizer," normalizing input files to the 16kHz mono format required by the AI model.
- **Docker Isolation:** The inference engine was containerized using **Docker**, ensuring that the AI environment remained isolated from the main application server, preventing resource contention and improving system stability.

3.5.4 Database and Cloud Integrations

The platform leverages a hybrid storage strategy to balance performance with data sovereignty.

- **MongoDB and Mongoose:** All user metadata and transcription histories are stored in **MongoDB**, with Mongoose providing a structured object-modeling layer.
- **Cloudinary and Brevo:** I integrated **Cloudinary** for the permanent storage of user profile images and voice metadata. For automated communication, I utilized the **Brevo REST API** to handle OTP (One-Time Password) verification and account security notifications.

3.5.5 DevOps and Deployment Pipeline

A professional CI/CD pipeline was established to maintain high project velocity and deployment reliability.

- **GitHub and Vercel: GitHub** served as the central version control hub, while **Vercel** was used for high-speed frontend hosting.
- **Render and Docker:** The backend services were deployed on **Render**, utilizing Docker containers to ensure that the complex environment required for AI inference remained consistent across staging and production environments.

Chapter Four

Technical Challenges and Solutions

Chapter Four: Technical Challenges and Solutions

The development of **VerbaSense** was not without significant technical hurdles. Building a local AI-powered platform required solving complex problems related to hardware limitations, data security and high-concurrency processing. This chapter details the primary challenges encountered during the internship and the engineering strategies implemented to overcome them.

4.1 Challenges Faced and Solutions

Throughout the project lifecycle, I encountered several "bottlenecks" that required a combination of research, experimentation and collaborative problem-solving with my supervisor.

4.1.1 Large File Upload Handling

The Challenge: Early in the development phase, the Express.js backend frequently crashed when users attempted to upload audio files larger than 10MB. The default memory limits of Node.js were insufficient to handle multiple simultaneous large-file uploads, leading to "Heap Out of Memory" errors. **The Solution:** I solved this by implementing a two-pronged approach. First, I adjusted the Express payload limits and configured **Multer** to use disk-based storage instead of memory-based buffering. This ensured that large files were streamed directly to the server's temporary storage, bypassing the RAM bottleneck. Second, I implemented a client-side validation layer that warned users about file size limits before the upload process began, significantly improving the system's stability.

4.1.2 Model Inference Latency

The Challenge: Initial tests using the Whisper "Large" model on standard CPU hardware resulted in unacceptable latency—sometimes taking up to five times the duration of the audio itself to produce a transcript. For a production-ready application, this was a significant barrier to user adoption. **The Solution:** After benchmarking different configurations in our technical labs, I moved the inference engine to the **graphics card** using the C++ optimized version of

Whisper. I also implemented a dynamic model-selection strategy where the "Base" model was used for quick previews, while the more intensive models were reserved for high-accuracy requests. This optimization reduced transcription time by over 70%, making the platform feel significantly more responsive.

4.1.3 Audio Format Incompatibility

The Challenge: The OpenAI Whisper engine is highly sensitive to input formats, requiring specific 16kHz mono WAV files. However, users frequently uploaded a wide variety of formats, including MP3, FLAC and high-bitrate stereo WAVs, which often caused the AI engine to fail or produce distorted transcripts. **The Solution:** I integrated a robust audio pre-processing pipeline using **FFmpeg**. I developed an automated script that intercepted every uploaded file and performed a "sanitization" pass—automatically converting it to the required 16kHz mono PCM format. This ensured that the AI engine received a consistent, high-quality stream every time, regardless of the user's original recording quality or format.

4.1.4 Secure Session Management

The Challenge: Because VerbaSense handles private and sensitive audio data, ensuring secure session management was paramount. Traditional local-storage-based authentication was deemed too risky due to potential XSS (Cross-Site Scripting) vulnerabilities. **The Solution:** I implemented a robust security architecture using **JSON Web Tokens (JWT)** stored exclusively in **HTTP-only cookies**. This approach prevents client-side scripts from accessing the session token, effectively mitigating the risk of token theft. Additionally, I configured a strict **CORS (Cross-Origin Resource Sharing)** policy to ensure that only authorized frontend domains could interact with our transcription API.

4.1.5 Resource Optimization for Local Deployment

The Challenge: During peak usage simulations, multiple concurrent transcription requests would often "max out" the server's CPU and VRAM, causing the entire system to hang or drop active connections. **The Solution:** I implemented a sophisticated **Task Queue** system using a producer-consumer pattern. Instead of processing every request immediately, the system now queues incoming tasks in a MongoDB-backed manager. A specialized "Worker" process then picks up these tasks sequentially based on the available system resources. This prevents system overloads and ensures that even during high-traffic periods, every user eventually receives their transcript with 100% accuracy.

Chapter Five

Professional Development

Chapter Five: Professional Development

My internship at USSBD served as a transformative bridge between academic theory and professional practice. Beyond the technical code, I gained a profound understanding of what it means to be a professional engineer in a high-stakes corporate environment. This chapter reflects on the diverse set of skills I acquired and the personal growth I experienced during my tenure.

5.1 Skills Gained and Learning Outcomes

The primary outcome of this internship was the successful transition from a classroom-based understanding of software to an industry-standard development mindset. I learned to navigate the entire software development lifecycle (SDLC), from initial requirement gathering and feasibility research to the final deployment of a production-ready application. One of the most significant learning outcomes was the realization that high-quality software is not just about writing functional code, but about ensuring security, scalability and maintainability in a real-world server environment.

5.2 Technical Skills Acquired

The technical demands of the VerbaSense project forced me to master a variety of modern software engineering frameworks and methodologies.

5.2.1 Full-Stack Development with Next.js 14

One of the most significant technical milestones was mastering the **Next.js 14** App Router architecture. I learned to balance server-side and client-side rendering to optimize performance. Implementing dynamic routes, server actions and layout-based state management taught me how to build scalable, high-performance web applications that are ready for production.

5.2.2 AI Model Integration and Optimization

Prior to this internship, my experience with AI was largely theoretical. Working with the **OpenAI Whisper** engine taught me the practical complexities of deploying AI models locally. I gained hands-on experience in C++ optimization, GPU acceleration and inference benchmarking. I learned how to "fine-tune" the interaction between a high-level web server and a low-level AI engine, a skill that is increasingly valuable in the modern software landscape.

5.2.3 Database Architecture and Security

Designing the **MongoDB** schema for VerbaSense provided me with deep insights into NoSQL database management. I learned how to design flexible, high-performance schemas that can handle unstructured transcription data. Furthermore, implementing **JWT-based authentication** and secure cookie management significantly improved my understanding of web security patterns and data sovereignty.

5.3 Professional and Soft Skills Gained

Technical proficiency is only one part of being an engineer; the ability to collaborate and communicate effectively is equally critical.

5.3.1 Collaborative Problem Solving

Working in a professional office environment, especially in direct coordination with my supervisor, taught me the value of collaborative brainstorming. I learned that the best solutions often arise from discussing architectural bottlenecks with a mentor. This experience improved my ability to listen actively, accept constructive criticism and iterate on my designs based on expert feedback.

5.3.2 Technical Documentation and Reporting

I developed a newfound respect for professional documentation. My mentor at USSBD emphasized that "undocumented code is legacy code." Consequently, I spent significant time drafting clear README files, API documentations and regular progress reports. This disciplined approach ensured that the VerbaSense project remains maintainable and transparent for future developers.

5.4 Personal Growth and Career Outlook

The internship provided me with the confidence and clarity needed to plan my future career path in the technology industry.

5.4.1 Building Engineering Confidence

Taking full ownership of the VerbaSense project—from research to deployment—significantly boosted my engineering confidence. Successfully navigating complex technical failures, such as GPU memory leaks or backend crashes, taught me that perseverance and systematic troubleshooting are the keys to solving any engineering challenge.

5.4.2 Future Career Path in AI and Web Development

This experience has solidified my interest in the intersection of Web Development and Artificial Intelligence. I now have a clear vision of how I want to grow as a full-stack engineer who specializes in integrating local AI solutions. The professional networks and technical foundations built at USSBD will undoubtedly serve as the cornerstone of my future career.

5.5 Professional Ethics and Work Culture

Finally, the internship exposed me to the professional ethics and values that define a successful software firm.

5.5.1 Organizational Values at USSBD

I was deeply impressed by the culture of innovation and accountability at USSBD. The organization's focus on privacy-first AI development aligned with my personal values. I learned how a firm's technical decisions are often driven by its underlying organizational philosophy, such as the commitment to data sovereignty and user privacy.

5.5.2 Responsibility and Accountability

In a professional setting, every line of code has real-world implications. I learned to take full accountability for my contributions, ensuring that my implementations were not only functional but also secure and ethically sound. This sense of responsibility is perhaps the most valuable "professional skill" I will carry forward into my career.

Chapter Six

Self Assessment

Chapter Six: Self Assessment

Reflecting on my internship experience at USSBD, I believe I demonstrated consistent dedication, adaptability and a proactive approach to learning. I entered the program with foundational knowledge in web development, but this internship allowed me to apply those concepts in a high-pressure, real-world environment and significantly broadened both my technical and professional capabilities.

One of my key strengths during the internship was my willingness to take initiative. Whether it was researching the C++ implementation of Whisper, troubleshooting complex GPU memory leaks or exploring advanced FFmpeg filters for audio sanitization, I made a conscious effort to go beyond basic instructions. I frequently reviewed server logs, explored deep documentation and sought out additional resources to better understand the underlying technologies. This self-driven learning helped me become more confident in identifying and resolving practical engineering problems.

I also maintained strong consistency in task completion and documentation. By adhering to iterative milestones, providing regular updates to my supervisor and reviewing each session's outcomes, I ensured that my progress was transparent and traceable. This habit not only improved my own retention of new concepts but also allowed me to contribute meaningfully during technical reviews and architectural discussions.

While I became highly comfortable with Next.js architecture and local AI inference, I recognized a gap in my familiarity with more advanced topics such as large-scale Docker orchestration and the fine-tuning of transformer-based models for specific dialects. Though these topics were touched upon during my research, I believe further exposure and hands-on practice are needed to gain full confidence in deploying such complex systems in high-availability production environments.

From a professional standpoint, I adapted well to the formal culture at USSBD. I was punctual, respectful in communication and cooperative in my coordination with my supervisor. I actively participated in technical discussions, accepted feedback positively and supported the project's goals whenever possible. In summary, I believe I made the most of this internship

opportunity, growing not only in my technical proficiency but also in my ability to work within a structured, professional environment.

6.1 Observations

Throughout the internship, several key observations stood out that shaped my perspective on the software industry:

- **Integration of R&D and Practice:** The structured approach at USSBD provided a balanced exposure to both theoretical research and hands-on device-level configurations. The hands-on nature of the tasks allowed for a much deeper understanding of concepts learned in university, such as asynchronous process management and database normalization.

- **Professional Mentorship:** The professionalism and framework within the Khulna branch created a productive learning environment. Having a supervisor who was approachable and willing to guide me through complex issues significantly improved my troubleshooting confidence and architectural thinking.

- **The Value of Documentation:** It became clear that documentation and consistent communication play a vital role in managing large-scale software operations. Without clear technical logs, even the most advanced code can become unmanageable over time.

Chapter Seven

Conclusion

Chapter Seven: Conclusion

The development of the **VerbaSense** platform successfully met and exceeded the original objectives of creating a professional, secure and highly accurate local Speech-to-Text service. This project demonstrated that with the right combination of modern frameworks like **Next.js 14**, **Node.js** and optimized AI engines like **OpenAI Whisper**, it is possible to build enterprise-grade AI solutions that prioritize data sovereignty and user privacy without relying on third-party cloud providers.

This internship at USSBD has been a cornerstone of my professional development. It provided a unique environment where theoretical research into transformer-based AI models could be immediately applied to a high-concurrency web application. Navigating the technical challenges of GPU memory management, audio normalization and secure session handling has equipped me with a robust engineering toolkit that goes far beyond what is typically covered in a university curriculum.

Beyond the technical accomplishments, the experience taught me the invaluable "soft skills" of the corporate world. Working under the direct guidance of my supervisor helped me understand the importance of clear technical communication, iterative planning and professional accountability. I am now far more confident in my ability to take a complex software requirement from the "research phase" all the way to "production deployment."

In summary, my time at the Khulna branch of USSBD was both challenging and deeply rewarding. I am leaving this internship with a functional, production-ready application in **VerbaSense** and a clear, inspired vision for my future career as a software engineer. The lessons learned here—both in code and in professional conduct—will undoubtedly serve as a solid foundation for my future endeavors in the global technology landscape.